[Def. Let $f_n : X \to [-\infty, \infty]$ $(n \in \mathbb{N})$ be functions.
Define for each $x \in X$,

$$\left(\sup_{n\in\mathbb{N}} f_n\right)(x) \overset{\text{def}}{=} \sup_{n\in\mathbb{N}} f_n(x) \quad \text{and} \quad \left(\limsup_{n\to\infty} f_n\right)(x) = \limsup_{n\to\infty} f_n(x).$$

[Lemma. Let $f : X \to [-\infty, \infty]$ with X is equipped a σ-algebra $\mathcal{M}$.
If $f^{-1}[(a, \infty]] \in \mathcal{M}$ for every $a \in \mathbb{R}$, then f is measurable.

Proof. Let $\Omega := \{E \subseteq [-\infty, \infty] \mid f^{-1}[E] \in \mathcal{M}\}$. Given $a \in \mathbb{R}$, let (a_n) be a sequence such that $a_n < a$ for all $n \in \mathbb{N}$ and $a_n \to a$ as $n \to \infty$.
(Concretely, we can choose $a_n := a - 1/n$.)
By the assumption, for each n, $f^{-1}[(a_n, \infty]] \in \mathcal{M}$ so that $(a_n, \infty] \in \Omega$.
Therefore,

$$[-\infty, a) = \bigcup_{n=1}^{\infty} [-\infty, a_n] = \bigcup_{n=1}^{\infty} (a_n, \infty]^c \in \Omega$$

because Ω is also σ-algebra, and thus $f^{-1}[[-\infty, a)] \in \mathcal{M}$.
Moreover, given $a, b \in \mathbb{R}$, $(a, b) = [-\infty, b) \cap (a, \infty]$ so that

$$f^{-1}[(a,b)] = f^{-1}[[-\infty, b)] \cap f^{-1}[(a, \infty]] = \left(f^{-1}[[-\infty, b)]^c \cup f^{-1}[(a, \infty]]^c\right)^c \in \mathcal{M}.$$